to which the collapse is simultaneous across space, the one according to which the left and right measurements occur at the same moment. But picking out a special perspective creates significant tension with the egalitarian core of special relativity. Proposals have been made to circumvent this problem, but debate continues regarding which, if any, are successful.[20]

Thus, although the majority view holds that there is a harmonious coexistence, some physicists and philosophers consider the exact relationship between quantum mechanics, entangled particles, and special relativity an open question. It's certainly possible, and in my view likely, that the majority view will ultimately prevail in some more definitive form. But history shows that subtle, foundational problems sometimes sow the seeds of future revolutions. On this one, only time will tell.

What Are We to Make of All This?

Bell's reasoning and Aspect's experiments show that the kind of universe Einstein envisioned may exist in the mind, but not in reality. Einstein's was a universe in which what you do right here has immediate relevance only for things that are also right here. Physics, in his view, was purely local. But we now see that the data rule out this kind of thinking; the data rule out this kind of universe.

Einstein's was also a universe in which objects possess definite values of all possible physical attributes. Attributes do not hang in limbo, waiting for an experimenter's measurement to bring them into existence. The majority of physicists would say that Einstein was wrong on this point, too. Particle properties, in this majority view, come into being when measurements force them to—an idea we will examine further in Chapter 7. When they are not being observed or interacting with the environment, particle properties have a nebulous, fuzzy existence characterized solely by a probability that one or another potentiality might be realized. The most extreme of those who hold this opinion would go as far as declaring that, indeed, when no one and no thing is "looking" at or interacting with the moon in any way, *it is not there.*